SPECIALIZATIONS ARE WITNESSED BY VALUATION RINGS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4042-4056, following the statement at 4024-4040.

Allowed standard prerequisites (editorial).

- Generalizations are the image of Spec of the local ring. lemma-specialize-points (card L26403 this batch, cited at ~4045) with the discussion following lemma-characterize-points (cited at ~4047). Produces the ring maps O_{S,s} -> kappa(s') -> K.
- Domination by a valuation ring. Algebra, Lemma algebra-lemma-dominate (cross-chapter, cited at ~4051). Chooses A subset K dominating the image of O_{S,s}.
- Morphisms from an affine scheme of a local ring. lemma-morphism-from-spec-local-ring (card L26396, cited at ~4054). Turns O_{S,s} -> A into f: Spec(A) -> S.

Source reference. Original reference 00IA: focused-seeds/geometry/S9-schemes-points-specialize/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01J9: sources/stacks-schemes/pinned-original-source.tex, source lines 2682–2682. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01J6: sources/stacks-schemes/pinned-original-source.tex, source lines 2550–2550. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01J7: sources/stacks-schemes/pinned-original-source.tex, source lines 2612–2612. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $s' \rightsquigarrow s$ be a specialization of points of S. Then*

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *there exists a valuation ring A and a morphism $f : \mathrm{Spec}(A) \to S$ such that the generic point η of $\mathrm{Spec}(A)$ maps to s' and the special point maps to s, and*
(2) *given a field extension $K/\kappa(s')$ we may arrange it so that the extension $\kappa(\eta)/\kappa(s')$ induced by f is isomorphic to the given extension.*

Proof. Let $s' \rightsquigarrow s$ be a specialization in S, and let $K/\kappa(s')$ be an extension of fields. By Lemma 01J7 and the discussion following Lemma 01J9 this leads to ring maps $\mathcal{O}_{S,s} \to \kappa(s') \to K$. Let $A \subset K$ be any valuation ring whose field of fractions is K and which dominates the image of $\mathcal{O}_{S,s} \to K$, see Algebra, Lemma 00IA. The ring map $\mathcal{O}_{S,s} \to A$ induces the morphism $f : \mathrm{Spec}(A) \to S$, see Lemma 01J6. This morphism has all the desired properties by construction. □